

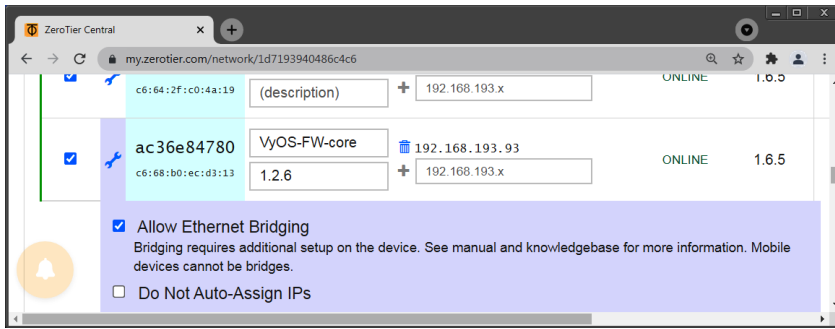


Figure 4: ZeroTier allows 'Ethernet bridging' per device

because ZeroTier is treated as a regular interface and can belong to a firewall zone.

ZeroTier Central comes with 'Flow Rules', which is an effective way to control the data streams from a central point. The configuration is less convenient; the firewall rules are created as text, similar to iptables rules. The resulting security policy applies to all ZeroTier clients connected to this network ID. As soon as the rule set is ready, the clients fetch the new policy and apply it locally.

The use of the firewall is recommended if incoming traffic originates from an untrusted location or if the ZeroTier switch is classified as insecure. In this case, the ZeroTier interface will receive a firewall policy that uses restrictive rules to block anything that is not explicitly allowed. The firewall policy can filter at both the IP layer and the Ethernet layer.

Self-hosting ZeroTier

The makers of ZeroTier maintain several root servers for all subscribers so that the client software can get in touch with the network and reach other nodes. It is possible to run one (better: two) root server and a network controller to independently build a ZeroTier cloud without outside dependencies. The required software is already installed because the 'ZeroTier One' client also functions as a network controller or root server.

Network controller

The first step to independence is to have your own network controller. ZeroTier uses the network controller as a configuration manager and certification authority. The configuration of the controller uses ZeroTier's API, which is well documented but inconvenient for daily use. Those who do not like to set up their new infrastructure with *curl*, *wget*, or *httplib* can use the Web interface from the third-party provider 'Key Networks'. The installation of the ZeroTier software and the Web UI is described in Listing 1 for a server running Debian Linux.

Listing 1: This script turns a Debian Linux host into a ZeroTier controller

```
wget http://download.zerotier.com/
debian/buster/pool/main/z/zerotier-one/
zerotier-one_1.8.1_amd64.deb
dpkg --install zerotier-one_1.8.1_
```

```
amd64.deb
systemctl restart zerotier-one
```

```
wget https://s3-us-west-1.amazonaws.
com/key-networks/deb/ztncai/1/x86_64/
ztncai_0.7.1_amd64.deb
dpkg --install ./ztncai_0.7.1_amd64.deb
echo 'HTTPS_PORT=8443' > /opt/key-
networks/ztncai/.env
systemctl restart ztncai
```

After the installation is complete, the Web interface is available at the URL <https://ipaddress:8443/>. The first login uses the credential admin/password. Subsequently, the Web UI forces a password change to boost security.

The configuration is very similar to the procedure from the previous section, although the Web UI of 'Key Networks' offers a smaller functionality than ZeroTier Central. The setup starts with the button 'Add network', which immediately creates a new network and presents the network ID. A client can directly join this network without knowing the network controller. The required command is no different from a regular 'join':

```
zerotier-cli join a0a9c4de8a520e4d
```

In the network controller's Web interface, the new members need authorisation and an IP address. In

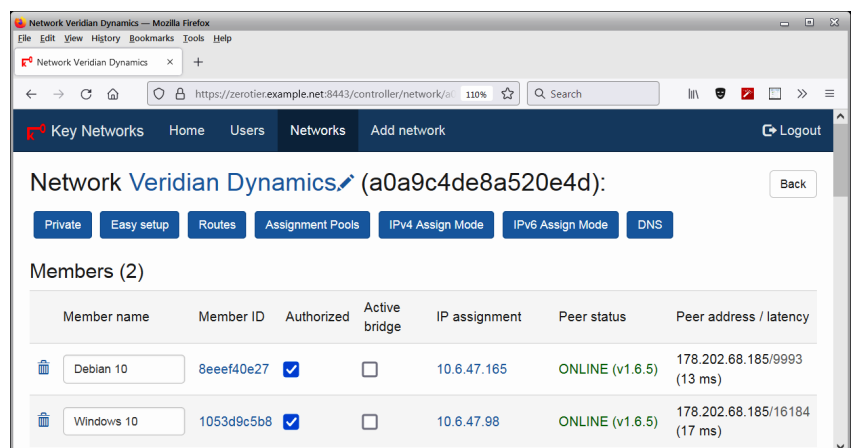


Figure 5: The clients have registered with the self-hosted network controller